

CTLRR-220PRO Long-range Millimeter-wave Radar

Product Introduction

CTLRR-220PRO is a 77GHz long-range millimeter-wave radar. It adopts an integrated new-generation RFCMOS domestic SoC solution, which can accurately measure the distance, speed and angle of target vehicles within 260 meters in front of the vehicle. It can also realize L2+ level intelligent auxiliary driving functions such as forward collision warning (FCW), automatic emergency braking (AEB), and adaptive cruise control system (ACC).

Radar appearance



Advantages

- *Integrating the latest-generation RFCMOS SOC, long and medium-range detection mode
- *Support dual-channel CAN-FD
- *Support 40 tracking target outputs
- *Full vehicle-grade design

*Support for applications such as FCW/AEB/ACC for medium and long-range use

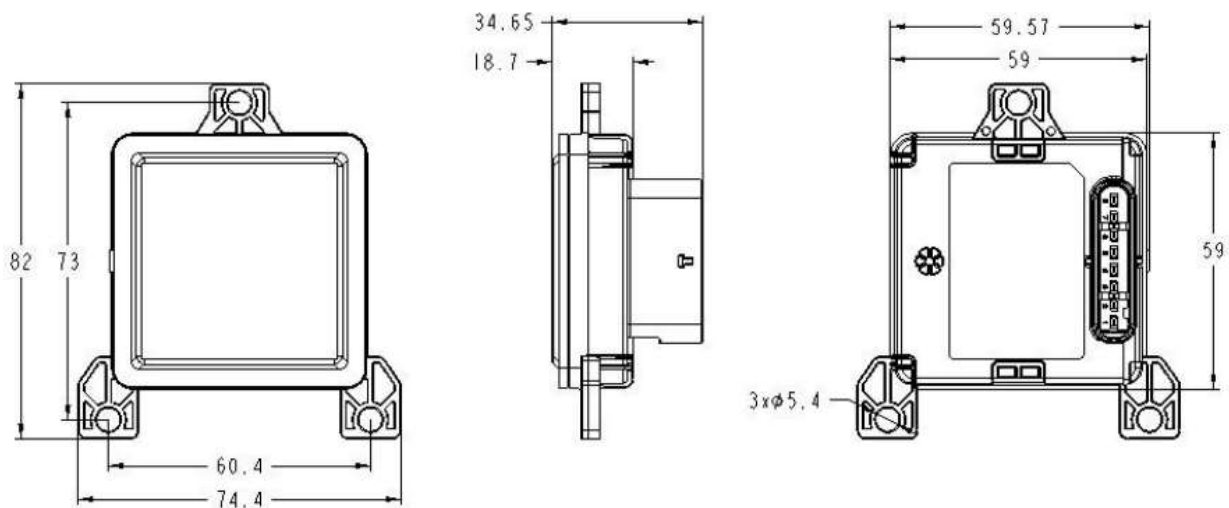
*Can be integrated with the visual ADAS system

Radar parameters

Items	Parameters
Frequency range	76 - 77 GHz
Equivalent Radiated Power (ERIP)	≤ 35 dBm
Measurement range	260m
Speed measurement range	-111m/s~+55.5m/s
Horizontal	$\pm 60^{\circ}$
Vertical	$\pm 15^{\circ}$
Distance measurement accuracy	± 0.1 m
Distance measurement accuracy	0.21m~0.5m
Horizontal angle accuracy	$\pm 0.1^{\circ}$
Horizontal angle resolution	2.5°
Pitch/roll angle accuracy	$\pm 0.2^{\circ}$
Speed measurement accuracy	± 0.03 m/s
Speed resolution	0.08m/s
Data interface	CAN/CAN FD
Operating voltage	9-16V DC

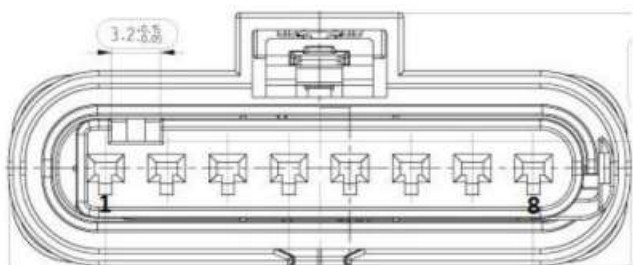
Overall power consumption	2.5W
Size	L59*W59*H18.7 mm
Weight	75g
Waterproof rating	IP6K7 & IP6K9K
Number of antenna channels	4T4R
Radar mode	FMCW
Refresh rate	50ms
Data model	Point Cloud/Track
Point cloud quantity	1024
Functional safety level	ASIL B

Structural dimensions (unit: mm)



Interface definition

Pin name	Signal Definition	Signal Function
1	KL15	Positive terminal of power supply
2	CAN1-H	CAN1 high (connected to the high voltage CAN of the vehicle)
3	CAN1-L	CAN1 low (connected to the low voltage CAN of the vehicle)
4	GND	Power ground
5	CAN2-H	CAN2 high (connected to the high voltage CAN of the MPC)
6	CAN2-L	CAN2 low (connected to the low voltage CAN of the MPC)
7	NC	Reserve
8	NC	Reserve



* 8-pin TYCO waterproof vehicle connector

* Model number: 1-1719393-1